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A Two-Phase Decomposition Method for Optimal Design of Looped Water Distribution Networks

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A two-phase decomposition method is proposed for the optimal design of new looped water distribution networks as well as for the parallel expansion of existing ones. The main feature of the method is that it generates a sequence of improving local optimal solutions. The first phase of the method takes a gradient approach with the flow distribution and pumping heads as decision variables and is an extension of the linear programming gradient method proposed by Alperovits and Shamir (1977) for nonlinear modeling. The technique is iterative and produces a local optimal solution. In the second phase the link head losses of this local optimal solution are fixed, and the resulting concave program is solved for the link flows and pumping heads; these then serve to restart the first phase to obtain an improved local optimal solution. The whole procedure continues until no further improvement can be achieved. Some applications and extensions of the method are also discussed.

INTRODUCTION

Mathematical models for the optimal design of looped water distribution networks and/or their parallel expansions are usually formulated as nonlinear programming (NLP) problems. Most of the published research in this area therefore explores efficient solution techniques which can search for a good local minimal solution or which are applicable to large-scale problems. Earlier examples of the NLP approaches described in the literature are variations of the Newton-Raphson method [e.g., *Shamir*, 1964; *Pitchai*, 1966; *Shamir and Howard*, 1968], penalty function methods [e.g., *Jacoby*, 1968; *Watanatada*, 1973], and the generalized reduced gradient method [*Shamir*, 1974]. Some other studies [e.g., *Bhave*, 1978, 1983, 1985; *Rowell and Barnes*, 1982] first select the branching configuration and identify the loop-forming links, then optimize the network system. However, the solution methods widely used decompose the original NLP problem into subproblems of smaller sizes and solve them alternately and iteratively.

One of the most significant decomposition methods is the linear programming gradient (LPG) method, proposed by *Alperovits and Shamir* [1977] (see also *Quindry et al.* [1979] and *Fujiwara et al.* [1987]). In this method, link flows are initially specified, and the resulting LP problem is solved for the optimal nodal heads and the lengths of pipe segments whose diameters are taken from commercially available sizes. The optimal dual variables for this LP problem help modify the assumed link flows to achieve a reduction in the system cost. A new LP problem with this modified link flows is then formulated and solved, and the whole process is repeated until a local optimal solution is obtained. *Quindry et al.* [1981] proposed a similar decomposition method, taking the nodal heads as the decision variables. They used the LP formulation by *Schaake and Lai* [1969], based on a piecewise linear approximation of a concave cost function with continuous pipe sizes.

A further advancement of the decomposition approach is given by *Mahjoub* [1983], who avoids the LP approximation by solving alternatively convex and concave programs as follows: first, link flows are fixed, and the resulting convex program (i.e., convex minimization over a convex set) is solved for link head losses, using the classical gradient projection method [e.g., *Rosen*, 1960]. Then the obtained optimal link head losses being fixed, the resulting concave program (i.e., concave minimization over a convex set) is solved for link flows. The whole procedure is repeated until no improvement can be attained. This decomposition scheme, however, does not assure that the solution obtained is locally optimal.

It should be noted that in the literature there is no method that allows further improvement once a local optimal solution is obtained. In contrast, a new two-phase iterative decomposition method is proposed in this paper, which generates a sequence of improving local optimal solutions. In the first phase the nonlinear programming gradient (NLPG) method is introduced, extending the LPG method by *Alperovits and Shamir* [1977]. The NLPG method first specifies the flow distribution and pumping heads and solves the resulting convex program for link head losses. The Lagrange multipliers of the optimal solution obtained are used to modify the flow distribution and pumping heads to achieve a reduction in system cost. A new convex program is then formulated and solved. The process is iterative and converges to a local optimal solution. In the second phase the link head losses of the local optimal solution obtained in the first phase are fixed, and the resulting concave program is solved for the flow distribution and pumping heads. This new optimum will serve as an initial solution to restart the first phase to obtain an improved local optimal solution. This two-phase iteration will then continue until no better local optimal solution is found. The proposed method therefore is an amalgam of the approaches proposed by *Alperovits and Shamir* [1977], *Quindry et al.* [1981], and *Mahjoub* [1983]. Since the subproblems in both phases are of smaller size than that of the original problem and their structure is also simpler (being a minimization of a convex or concave objective function subject to linear constraints), the method is computationally superior to the direct application of general purpose NLP codes. Although this new algorithm cannot guarantee the global optimality of the obtained solution, it allows a move from one local optimal solution to a

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different and better one which may be located far away, and thus it can be considered as an approach toward the global optimal solution. The method is illustrated through the planned water distribution trunk network in Hanoi, Vietnam.

The paper also considers the problem of optimally expanding an existing water distribution network where additional pumping facilities and new pipe installation are considered. It will be shown that although the involvement of the parallel pipes and the pumping costs makes the first phase subproblem a nonconvex program for link head losses, the NLPG method remains valid for finding a local optimal solution. Since the second-phase subproblem is always a concave program, the two-phase method proposed can also be applied to this problem. In fact, when applied to the problem of the New York City water supply system [*Schaake and Lai*, 1969], the solution obtained by the new method is better than any solution so far presented in the literature.

MATHEMATICAL MODEL OF NEW DESIGN PROBLEM

Consider a water distribution network represented by a connected and directed graph with n nodes, t links, and m independent and oriented loops. For simplicity and clarity of presentation only networks with a single source (node 1), a single demand pattern, and a new pumping facility at the source node are considered. More complicated situations such as multiple sources and multiple loading patterns will be discussed later. Let L_i be the length of link i, and let J_i and q_i be the head loss and flow rate along link i. Let a_k and $h_k^{\min}$ denote the water demand and minimum required hydraulic head at node k; and let H_1 denote the head lift provided by pumping at the source node 1.

The diameter of the pipe along link i, denoted by x_i ($i = 1, \cdots, t$), is taken as a continuous variable. Following *Schaake and Lai* [1969], *Swamee et al.* [1973], *Quindry et al.* [1981], and *Bhave* [1985], the total network design cost is expressed as a function of pipe diameters and pump heads:

$$f(x_1, \cdots, x_t, H_1) = b_1 \sum_i L_i x_i^{\beta} + b_2 |a_1|^{e_1} H_1^{e_2} + b_3 |a_1| H_1 \qquad (1)$$

where the first term is the cost of pipes, the second is the pumping capital cost, and the last is the pump operating cost; $-a_1$ is the supply water flow rate at the source node 1; and b_1, b_2, b_3, e_1, e_2, and β are constants. Note that $0 < e_1$, $e_2 < 1$ and $1 < \beta < 2.5$ [see *Swamee et al.*, 1973]. The problem is to find the pipe diameter x_i, pumping head H_1, flow rate q_i, and head loss J_i in each link i to minimize the objective function (1) under the following constraints:

1. Flow balance at each node

$$\sum_{i \in \mathrm{in}(k)} q_i - \sum_{i \in \mathrm{out}(k)} q_i = a_k \qquad k = 1, \cdots, n \qquad (2)$$

where a_k is the demand at node k, and in(k) and out(k) are the set of all links connected into and out of node k, respectively. Note that $a_k > 0$ if k is a demand node, and $a_k < 0$ if k is a supply node.

2. Head loss balance along each loop

$$\sum_{i \in \mathrm{loop}\ p} \pm J_i = 0 \qquad p = 1, \cdots, m \qquad (3)$$

where the summation is taken along the basic loop p, and J_i takes a positive sign if the orientation of loop p and the flow direction coincide in link i; otherwise J_i takes a negative sign.

3. The hydraulic head requirement at each node

$$\sum_{i \in r(k)} \pm J_i \leqq H_1 + h_1 - h_k^{\min} \qquad k = 1, \cdots, n \qquad (4)$$

where h_1 is the given initial hydraulic head without pumping at the source node 1 and $r(k)$ is a directed path connecting this source node with node k. Here also, J_i takes a positive sign if the path $r(k)$ and the flow direction coincide in link i; otherwise it takes a negative sign.

4. Bounding constraints

$$q_i \geqq q_i^{\min} \qquad i = 1, \cdots, t \qquad (5)$$

$$H_1 \geqq 0 \qquad (6)$$

$$d_{\max} \geqq x_i \geqq d_{\min} \qquad i = 1, \cdots, t \qquad (7)$$

where $q_i^{\min}$ is the minimum required flow rate along link i, and $d_{\max}$ and $d_{\min}$ are upper and lower bounds for diameter of pipes.

5. Hazen-Williams equations

$$J_i = KL_i(q_i/c)^{\lambda} x_i^{-\alpha} \qquad i = 1, \cdots, t \qquad (8)$$

where $\lambda = 1.85$ and $\alpha = 4.87$ (thus $1/\lambda \approx 0.54$ and $\alpha/\lambda \approx 2.63$), while K and c are Hazen-Williams coefficients whose values depend on the measurement units and the characteristics of the pipe.

Eliminating the decision variable x_i from the Hazen-Williams equation (8) and substituting it in the objective function (1) and the bounding constraint (7), the problem can be restated as follows:

minimize

$$f(q, H_1, J) = b_1 \sum_i L_i(KL_i/J_i)^{\beta/\alpha}(q_i/c)^{\beta\lambda/\alpha} + b_2|a_1|^{e_1} H_1^{e_2} + b_3|a_1|H_1 \qquad (9)$$

subject to

$$\sum_{i \in \mathrm{in}(k)} q_i - \sum_{i \in \mathrm{out}(k)} q_i = a_k \qquad k = 1, \cdots, n$$

$$\sum_{i \in \mathrm{loop}\ p} \pm J_i = 0 \qquad p = 1, \cdots, m$$

$$\sum_{i \in r(k)} \pm J_i \leqq H_1 + h_1 - h_k^{\min} \qquad k = 1, \cdots, n$$

$$q_i \geqq q_i^{\min} \qquad i = 1, \cdots, t$$

$$H_1 \geqq 0$$

$$J_i \leqq KL_i(q_i/c)^{\lambda}/d_{\min}^{\alpha} \qquad i = 1, \cdots, t \qquad (10)$$

$$KL_i(q_i/c)^{\lambda}/d_{\max}^{\alpha} \leqq J_i \qquad i = 1, \cdots, t \qquad (11)$$

The decision variables in this problem are pumping pressure H_1, flow rates $q = [q_1, \cdots, q_t]$ and head losses $J = [J_1, \cdots, J_t]$. Note that since $0 < e_2$, $\lambda\beta/\alpha$, $\beta/\alpha < 1$, the

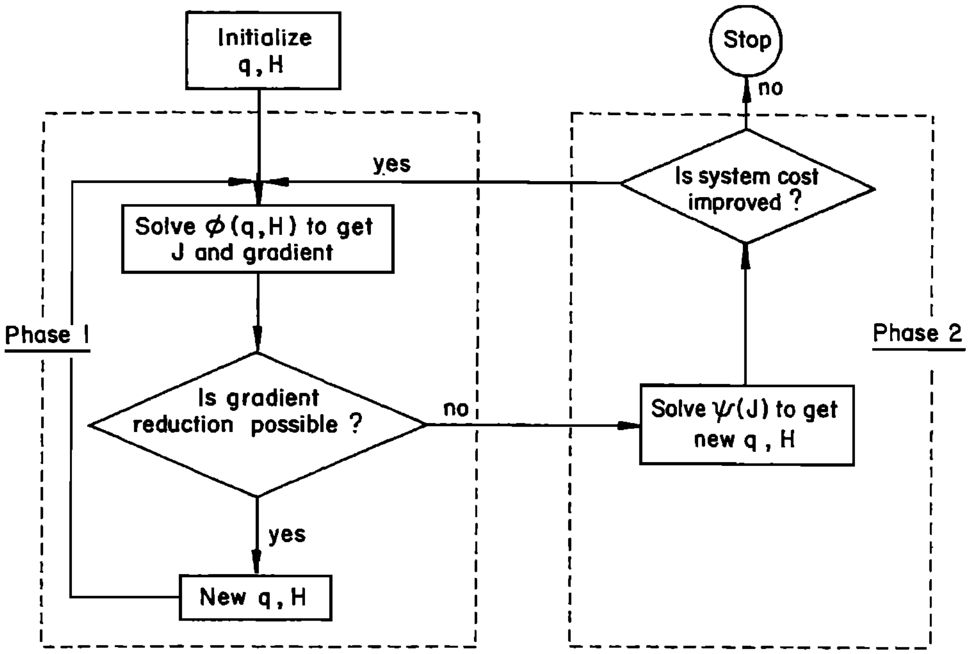

Fig. 1. Two-phase decomposition method.

objective function $f(q, H_1, J)$ is convex in J and concave in q and H_1.

Two-Phase Decomposition Method

The method proposed in this paper is an iterative procedure, each step of which is divided into two phases. In the first phase a new method, the nonlinear programming gradient (NLPG) method, is proposed which is an extension of the LPG method of *Alperovits and Shamir* [1977] and *Quindry et al.* [1979]. This method takes the link flows and the pumping heads as decision variables and then a gradient is calculated to modify the link flows as well as the pumping heads to obtain a reduction in the system cost. However, with the NLPG method, the gradient is computed using the optimal Lagrange multipliers of an NLP solution, whereas the dual solution of an LP is used in the standard LPG method. It will be proved that this phase produces a local optimal solution to the original problem. This solution is then used to formulate a concave programming problem in the second phase by fixing the head losses along the links. By solving this concave program, a new and better feasible solution can be obtained which will serve as an initial solution to restart the first phase of the next iteration. The whole procedure then continues until no improvement can be achieved. The flow chart of the method is given in Figure 1. The two phases will be described in the following subsections.

It can be noted that *Mahjoub* [1983] uses a similar decomposition scheme. In fact, he simply fixes q and solves the convex program for J and then fixes this optimal J to solve the concave program for q. Thus although Mahjoub repeats this iteration until the two solutions converge, his algorithm does not necessarily guarantee the local optimality, in both variables q and J, of the obtained solution.

First Phase: Nonlinear Programming Gradient (NLPG) Method

Let $\bar q = [\bar q_1, \cdots, \bar q_t]$ be any feasible flow distribution (i.e., $\bar q_i$ satisfies the flow balance equations (2) and the bounding constraints (5)) and $\bar H_1$ be any nonnegative value of pumping head. The problem with fixed $\bar q$ and $\bar H_1$, denoted by $\phi(\bar q, \bar H_1)$, is formulated as follows:

minimize

$$f(\bar q, \bar H_1, J) = b_1 \sum_i L_i(KL_i/J_i)^{\beta/\alpha}(\bar q_i/c)^{\beta\lambda/\alpha} + b_2|a_1|^{e_1}\bar H_1^{e_2} + b_3|a_1|\bar H_1 \quad (12)$$

subject to

$$\sum_{i \in \text{loop } p} \pm J_i = 0 \qquad p = 1, \cdots, m$$

$$\sum_{i \in r(k)} \pm J_i \leqq \bar H_1 + h_1 - h_k^{\min} \qquad k = 1, \cdots, n \quad (13)$$

$$J_i \leqq KL_i(\bar q_i/c)^{\lambda}/d_{\min}^{\alpha} \qquad i = 1, \cdots, t \quad (14)$$

$$KL_i(\bar q_i/c)^{\lambda}/d_{\max}^{\alpha} \leqq J_i \qquad i = 1, \cdots, t \quad (15)$$

It should be noted here that $\phi(\bar q, \bar H_1)$ is a convex program since the objective function is convex and all the constraints are linear in J, thus any local optimal solution is also a global one, and efficient minimization techniques can be applied.

Let $\Phi(\bar q, \bar H_1)$ be the optimal value of $\phi(\bar q, \bar H_1)$, $\bar J = [\bar J_1, \cdots, \bar J_t]$ be the optimal solution of $\phi(\bar q, \bar H_1)$, and $\bar u = [\bar u_1, \cdots, \bar u_n]$, $\bar v = [\bar v_1, \cdots, \bar v_t]$, and $\bar w = [\bar w_1, \cdots, \bar w_t]$ be the optimal Lagrange multipliers corresponding to the constraints (13), (14), and (15), respectively. Then, assuming sufficient conditions for the nonlinear programming sensitivity theorem (namely, the second-order sufficient conditions for a local optimum of $\phi(\bar{q}, \bar{H}_1)$, the linear independency of the gradients for the active constraints, and the strict complementarity slackness conditions of $\phi(\bar{q}, \bar{H}_1)$ at $(\bar{J}, \bar{u}, \bar{v}, \bar{w})$ and the twice continuous differentiability of the functions defining $\Phi(q, H_1)$ in (q, H_1, J) near $(\bar{q}, \bar{H}_1, \bar{J})$ [*Fiacco*, 1983, theorems 3.2.2 and 3.4.1]), the partial derivative of Φ, as a function of q and H_1, with respect to q_i $(i = 1, \cdots, t)$ and H_1 evaluated at $q = \bar{q}$ and $H_1 = \bar{H}_1$ is expressed as

$$\left.\frac{\partial \Phi(q, H_1)}{\partial q_i}\right|_{\substack{q=\bar{q}\\ H_1=\bar{H}_1}} = \left.\frac{\partial f(q, H_1, \bar{J})}{\partial q_i}\right|_{\substack{q=\bar{q}\\ H_1=\bar{H}_1}}$$

$$- \sum_{s=1}^{t} \bar{v}_s \frac{\partial}{\partial q_i}\left[\frac{Kc^{-\lambda}q_s^{\lambda}L_s}{d_s^{\alpha}}\right]\Bigg|_{\substack{q=\bar{q}\\ H_1=\bar{H}_1}}$$

$$+ \sum_{s=1}^{t} \bar{w}_s \frac{\partial}{\partial q_i}\left[\frac{Kc^{-\lambda}q_s^{\lambda}L_s}{(d_{\max}^{\alpha/\lambda} + d_s^{\alpha/\lambda})^{\lambda}}\right]\Bigg|_{\substack{q=\bar{q}\\ H_1=\bar{H}_1}}$$

$$= b_1 L_i[(KL_i/\bar{J}_i)^{\beta/\alpha} c^{-\beta\lambda/\alpha}(\beta\lambda/\alpha) q_i^{\beta\lambda/\alpha - 1}$$

$$- \bar{v}_i[Kc^{-\lambda}L_i/d_i^{\alpha}]\lambda \bar{q}_i^{\lambda-1} + \bar{w}_i[Kc^{-\lambda}L_i/d_{\max}^{\alpha}]\lambda \bar{q}_i^{\lambda-1} \quad (16)$$

$$\left.\frac{\partial \Phi(q, H_1)}{\partial H_1}\right|_{\substack{q=\bar{q}\\ H_1=\bar{H}_1}}$$

$$= \bar{H}_1 q = \bar{q} = b_2|a_1|^{e_1} e_2 \bar{H}_1^{e_2-1} + b_3(-a_1) - \sum_{k=1}^{n} \bar{u}_k \quad (17)$$

In order to reduce the problem size, variables q_i $(i = 1, \cdots, t)$ can be replaced by new variables Δ_p $(p = 1, \cdots, m)$, the flow change along the orientation of each loop p as shown below. The reduced problem size is due to the fact that for any plane network $m = t - (n - 1)$, which is in general much smaller than n [e.g., *Berge*, 1985]. Note that the flow change in link i can be expressed by $\Sigma_{p=1}^{m} \sigma_{ip}\Delta_p$, where $\sigma_{ip} = +1$ if link i lies on loop p and the flow change direction coincides with the orientation of loop p in link i; $\sigma_{ip} = -1$ if link i lies on loop p and the flow change direction is opposite to the orientation of loop p in link i; otherwise $\sigma_{ip} = 0$. Now for the given flow distribution $\bar{q}$ any feasible flow distribution q is uniquely expressed as

$$q_i = \bar{q}_i + \sum_{p=1}^{m} \sigma_{ip}\Delta_p \qquad i = 1, \cdots, t \quad (18)$$

for some loop flow changes $\Delta_1, \cdots, \Delta_m$. Thus $\partial\Phi/\partial\Delta_p$ can be expressed as

$$\partial\Phi/\partial\Delta_p = \sum_{i=1}^{t} (\partial\Phi/\partial q_i)(\partial q_i/\partial\Delta_p)$$

$$= \sum_{i=1}^{t} \sigma_{ip}\partial\Phi/\partial q_i \qquad p = 1, \cdots, m \quad (19)$$

These formulas (16)–(19) are similar to the gradient expression of the LPG method [see *Alperovits and Shamir*, 1977; *Quindry et al.*, 1979; *Fujiwara et al.*, 1987]. To reduce the system cost the flow distribution $\bar{q}$ is modified along each loop, together with the pumping head $\bar{H}_1$, following the modified LPG approach proposed by *Fujiwara et al.* [1987], where the search direction is determined by a quasi-Newton method with Broyden-Fletcher-Goldfarb-Schanno secant update, and the step size is determined by a backtracking line search strategy [*Dennis and Schnabel*, 1983]. With this new flow distribution $\hat{q}$ and pumping pressure $\hat{H}_1$ the resulting problem $\phi(\hat{q}, \hat{H}_1)$ is again solved. This iterative procedure stops when either the gradient is zero at some iteration or the line search fails to produce significant flow changes to reduce the system cost in the assigned direction. As shown in the appendix, the solution obtained is a local optimal solution of the original problem.

Second Phase: Improving Local Optimal Solution

Suppose a local optimal solution $(q^{(1)}, H_1^{(1)}, J^{(1)})$ has been obtained by using the NLPG method described above. A better local optimal solution $(q^{(2)}, H_1^{(2)}, J^{(2)})$ can be obtained in the following manner. First, we fix $J^{(1)}$ and consider the following problem, $\psi(J^{(1)})$, over q and H_1:

minimize

$$f(q, H_1, J^{(1)}) = b_1 \sum_i L_i(KL_i/J_i^{(1)})^{\beta/\alpha}(q_i/c)^{\beta\lambda/\alpha}$$

$$+ b_2|a_1|^{e_1}H_1^{e_2} + b_3|a_1|H_1 \quad (20)$$

subject to

$$\sum_{i\in \mathrm{in}(k)} q_i - \sum_{i\in \mathrm{out}(k)} q_i = a_k \qquad k = 1, \cdots, n$$

$$\sum_{i\in r(k)} \pm J_i^{(1)} \leqq H_1 + h_1 - h_k^{\min} \qquad k = 1, \cdots, n \quad (21)$$

$$q_i \geqq q_i^{\min} \qquad i = 1, \cdots, t$$

$$H_1 \geqq 0$$

$$q_i \geqq (J_i^{(1)}/KL_i)^{1/\lambda} c d_{\max}^{\alpha/\lambda} \qquad i = 1, \cdots, t \quad (22)$$

$$q_i \geqq (J_i^{(1)}/KL_i)^{1/\lambda} c d_{\min}^{\alpha/\lambda} \qquad i = 1, \cdots, t \quad (23)$$

Note that $\psi(J^{(1)})$ is a concave program, since the objective function is concave and all the constraints are linear in q and H_1. This concave minimization problem has been studied extensively over the past two decades [e.g., *Tuy*, 1964, 1983, 1987; *Zwart*, 1974; *Falk and Hoffman*, 1976; *Thieu et al.*, 1983]. Since the basic property of this problem is that a global solution can always be found at one of the vertices of the feasible region [see *Tuy*, 1964], a natural approach to the solution is to find all the vertices and compare the objective functions evaluated at these vertices.

The global optimal solution for $\psi(J^{(1)})$ can thus be obtained, serving as an initial solution to restart the NLPG method and to find a new local optimal solution $(q^{(2)}, H_1^{(2)}, J^{(2)})$. Since during both the concave programming phase and the NLPG method the objective value decreases, it is clear that $(q^{(2)}, H_1^{(2)}, J^{(2)})$ is a better local optimal solution than $(q^{(1)}, H_1^{(1)}, J^{(1)})$.

MODEL FOR PARALLEL EXPANSION OF EXISTING NETWORKS

Consider a situation in which existing facilities cannot satisfy all the requirements of water quantity and pressure at

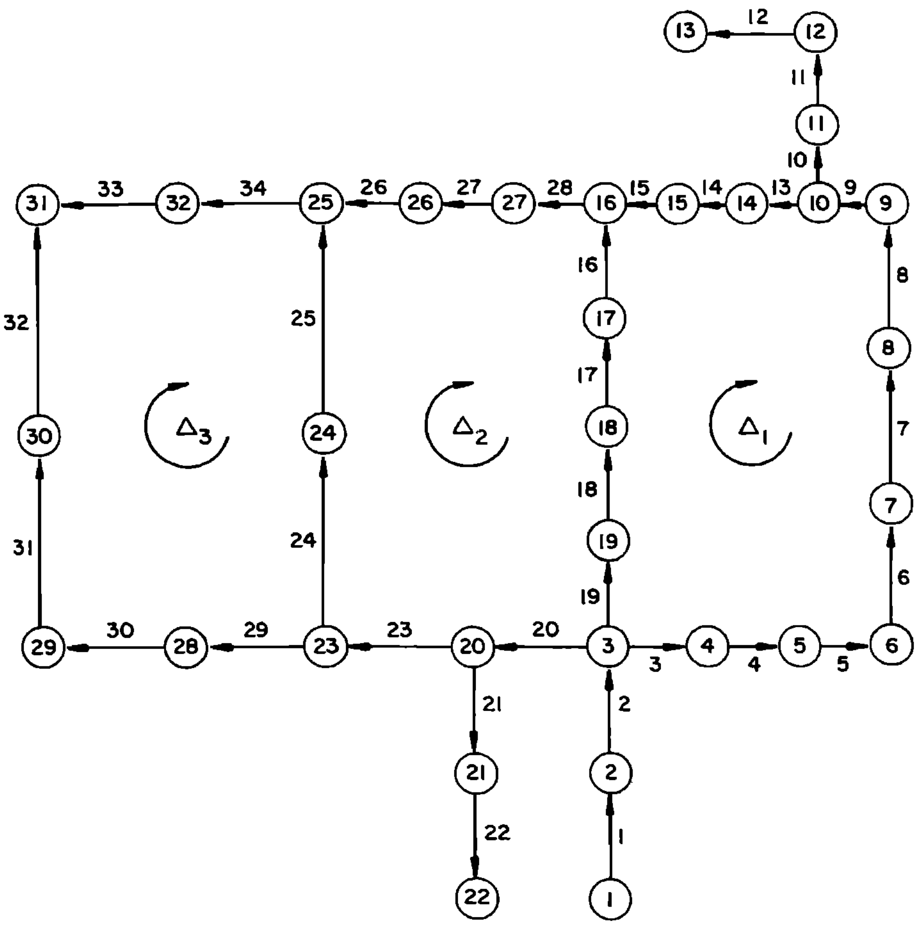

Fig. 2. Hanoi water distribution network.

the demand nodes; a decision is made to install new pipes parallel to the existing ones and to add new pumping facilities to meet these requirements. The problem is to determine the diameters of the new pipes and the capacity of the new pumping facilities so that the total cost of expansion (the cost of the new pipes and the capital and operation cost of new pumping facilities) is minimized.

In formulating the mathematical model for this problem the same assumptions as in the previous model are presumed, and the same symbols are adopted except that x_i now stands for the diameter of the new parallel pipe in link i, and $q_i = q_{i,O} + q_{i,N}$, where $q_{i,O}$ and $q_{i,N}$ are the flow rates in the old and new pipe of link i, respectively. Let d_i denote the existing pipe diameter of link i. Then the objective function (1) and the constraints (2)–(5) remain unchanged, while the constraints (7) become

$$d_{\max} \geqq x_i \geqq 0 \qquad i = 1, \cdots, t \tag{24}$$

where $x_i = 0$ implies no new pipe installation in link i. The Hazen-Williams equation (8) is replaced by

$$J_i = KL_i(q_{i,O}/c)^{\lambda} d_i^{-\alpha} = KL_i(q_{i,N}/c)^{\lambda} x_i^{-\alpha} \tag{25}$$

which can now be written as

$$q_i = c(J_i/KL_i)^{1/\lambda}(d_i^{\alpha/\lambda} + x_i^{\alpha/\lambda}) \tag{26}$$

Eliminating x_i from (26) and substituting it in the objective function (1), the first term of (1) becomes

$$b_1 \sum_i L_i[(KL_i/J_i)^{1/\lambda} q_i/c - d_i^{\alpha/\lambda}]^{\beta\lambda/\alpha} \tag{27}$$

while the bounding constraints (24) become

$$J_i \leqq KL_i(q_i/c)^{\lambda}/d_i^{\alpha} \qquad i = 1, \cdots, t \tag{28}$$

$$KL_i(q_i/c)^{\lambda}/(d_{\max}^{\alpha/\lambda} + d_i^{\alpha/\lambda})^{\lambda} \leqq J_i \qquad i = 1, \cdots, t \tag{29}$$

Note that although $f(\bar{q}, \bar{H}_1, J)$ is not convex in J anymore, all the constraints are still linear in J, thus a simple and efficient minimizing method such as the active set method [see *Gill et al.*, 1981; *Fletcher*, 1987] can be applied to the subproblem $\phi(\bar{q}, \bar{H}_1)$. The modified calculation of the gradients $\partial\Phi(q, H_1)/\partial q_i$ and $\partial\Phi(q, H_1)/\partial H_1$ are straightforward. Since $f(q, H_1, J)$ remains concave in q and H_1, the two-phase method can be applied to this problem.

APPLICATION TO DESIGN OF NEW WATER NETWORKS

As an application of the proposed algorithm, the planned water distribution trunk network in Hanoi, Vietnam, is considered. The configuration of the network, consisting of 32 nodes, 34 links, and 3 loops, is given in Figure 2. No

TABLE 1. Data Set for Hanoi Water Distribution Network

Link	Length, m	Initial flow, $m^3\ h^{-1}$	Node	Minimum Head Required, m	Initial Head, m
1	100	19940	1	100	100
2	1350	19050	2	30	95
3	900	8705	3	30	90
4	1150	8575	4	30	86
5	1450	7850	5	30	82
6	450	6845	6	30	80
7	850	5495	7	30	75
8	850	4945	8	30	72
9	800	4420	9	30	70
10	950	2000	10	30	65
11	1200	1500	11	30	60
12	3500	940	12	30	50
13	800	1895	13	30	40
14	500	1280	14	30	60
15	550	1000	15	30	55
16	2730	1105	16	30	50
17	1750	1970	17	30	60
18	800	3315	18	30	70
19	400	3375	19	30	80
20	2200	6120	20	30	80
21	1500	1415	21	30	60
22	500	485	22	30	40
23	2650	3430	23	30	70
24	1230	1320	24	30	60
25	1300	500	25	30	40
26	850	525	26	30	42
27	300	1425	27	30	45
28	750	1795	28	30	60
29	1500	1065	29	30	50
30	2000	775	30	30	40
31	1600	415	31	30	30
32	150	55	32	30	35
33	860	50			
34	950	855			

pumping facilities are considered. The link lengths, demands, and hydraulic heads at nodes are presented in Table 1. The minimum required flow rate is 5 m^3/hr and the upper and lower bounds for the pipe diameters are 40 and 12 inches respectively, as the set of commercially available diameters is 12, 16, 20, 24, 30, and 40 inches. Other necessary data regarding Hazen-Williams coefficients and cost coefficients are given as follows: $K = 162.5$, $c = 130$, $\alpha = 4.87$, $\beta = 1.5$, $b_1 = 1.1$, and $\lambda = 1.85$. The whole procedure is coded in FORTRAN and run on an AT compatible microcomputer in about 1 hour. The NPSOL Fortran Subroutine Library, release 4.1, of Stanford University is used to solve the convex program of phase one.

With the initial feasible flow pattern as given in Table 1, phase one terminates after 23 iterations, reducing the system cost from an initial value of 5.849 to 5.414 million dollars (that is, a reduction of 7.4% in system cost). Now solving the concave problem in the second phase, the global optimum $\Delta = [\Delta_1, \Delta_2, \Delta_3] = [689.69, 314.84, 0]$ is found, with the corresponding system cost equal to 5.358 million dollars, showing a reduction of 1.0%. These optimal flow changes give a new feasible flow pattern to restart phase one, which now terminates after six iterations. Now the phase two gives the optimum Δ to be [0, 0, 0], which means that one cannot improve further on the system cost by changing the flows while maintaining the nodal heads fixed. The whole procedure thus terminates with a final system cost of 5.354 million dollars; this is a total reduction of 8.5%. The final solution is presented in Table 2.

Note the magnitude of the change induced by phase two, which shows that the concave program allows a "global" move from the current local optimal solution. The final optimal diameters can also be converted to the discrete set of commercially available diameters using the standard split pipe technique. The results are summarized in Table 3, providing a realistic cost of 5.562 million dollars.

APPLICATION TO PARALLEL EXPANSION PROBLEM

In this section the NLPG method is applied to the parallel expansion problem of the New York City water supply system, which has been previously studied by several authors. The layout of the network is shown in Figure 3. The original data set is taken from *Quindry et al.* [1981] and is summarized in Tables 4 and 5, which also include the initial flow rates and the final solution obtained by our method. The following parameters are taken from *Schaake and Lai* [1969]: $b_1 = 1.1$, $\beta = 1.24$ while $K = (8.515)10^5$ (for length in feet and diameter in inches), and $c = 100$ [see *Quindry et al.*, 1981]. The nonlinear programming problem $\phi(\bar{q})$ is solved using the same NPSOL subroutine library.

The numerical results of the proposed method are summarized and compared with the results of previous authors [*Schaake and Lai*, 1969; *Quindry et al.*, 1981; *Bhave*, 1985;

TABLE 2. Final Optimal Solution for Hanoi Network

Link	Optimal Diameter, inches	Optimal Flow, $m^3\ h^{-1}$	Node	Minimum Head Required, m	Optimal Head, m
1	40.0	19,940	1	100	100.0
2	40.0	19,050	2	30	98.6
3	38.8	8,003	3	30	81.4
4	38.7	7,873	4	30	75.9
5	37.8	7,148	5	30	69.0
6	36.3	6,143	6	30	60.9
7	33.8	4,793	8	30	58.5
8	32.8	4,243	8	30	54.7
9	31.5	3,718	9	30	51.0
10	25.0	2,000	10	30	47.8
11	23.0	1,500	11	30	44.0
12	20.2	940	12	30	39.8
13	19.0	1,193	13	30	30.0
14	14.5	578	14	30	43.2
15	12.0	298	15	30	40.3
16	19.9	1,474	16	30	38.0
17	23.1	2,339	17	30	56.7
18	26.6	3,684	18	30	70.5
19	26.8	3,744	19	30	77.8
20	35.2	6,452	20	30	67.0
21	16.4	1,415	21	30	42.3
22	12.0	485	22	30	37.2
23	29.5	3,762	23	30	51.9
24	19.3	1,634	24	30	40.1
25	16.4	814	25	30	32.6
26	12.0	193	26	30	34.1
27	20.0	1,093	27	30	35.2
28	22.0	1,463	28	30	44.3
29	18.9	1,083	29	30	35.3
30	17.1	793	30	30	30.0
31	14.6	433	31	30	30.0
32	12.0	73	32	30	30.1
33	12.0	32			
34	19.5	837			

Morgan and Goulter, 1985] in Table 6, where all the costs are computed by the same cost function $\Sigma_i\ 1.1\ L_i\ d_i^{1.24}$ as given by *Schaake and Lai* [1969]. The number of new pipes with the different optimal solutions is reduced from 18 [*Schaake and Lai*, 1969] to 6 (*Bhave* [1985], *Morgan and Goulter* [1985], and in the present study). Bhave reduced the optimal cost of 77.611 million dollars given by Schaake and Lai to 40.180 million, while Morgan and Goulter were able to reduce the cost to 39.229 million dollars. In this study, costs have been reduced even further to 36.076 million dollars, which is a 6.2% reduction compared with the current best result of 39.229 million dollars by *Morgan and Goulter* [1985]. Moreover, all the six links used for new pipes in the study by *Morgan and Goulter* [1985] and the present study coincide, while five links coincide with those in the study by *Bhave* [1985].

Since *Morgan and Goulter* [1985] used preselected pipe diameters (in 12 inch intervals) rather than continuous ones, in order to make the comparison more valid, we converted our continuous solution to the same range of discrete diameters, using the standard split pipe technique, and compared it with the result given by Morgan and Goulter. The comparison is presented in Table 7, which still shows a reduction of 5.9% in the system cost.

It is observed that for this problem the concave program in phase two reduces the feasible region to a single point and therefore cannot indicate any improvement of the local optimal solution. The problem is thus only an example to demonstrate the NLPG method for the parallel expansion model. The problem, due to its small size, can be solved with an AT microcomputer in twenty minutes.

EXTENSION TO COMPLEX SYSTEMS

In this section the models and methods discussed above will be extended by introducing a number of new variables and constraints to incorporate more complex situations.

Multiple Sources

If the network has more than one, say z, source node, the pumping costs at each of these nodes can be added, giving the objective function as

$$b_1 \sum_{i=1}^{t} L_i x_i^{\beta} + b_2 \sum_{k=1}^{z} |a_k|^{e_1} H_k^{e_2} + b_3 \sum_{k=1}^{z} |a_k| H_k \quad (30)$$

where $-a_k$ is the supply flow rate and H_k is the additional pumping pressure at the source node k ($k = 1, \cdots, z$). Moreover, the following fixed pressure constraint will be added for each source node $k = 2, \cdots, z$:

$$\sum_{i \in r(1,k)} \pm J_i = H_1 + h_1 - (H_k + h_k) \quad (31)$$

TABLE 3. Conversion to Commercial Diameters

Link	Length, km	Optimal Diameter, inches	Realistic Diameter, inches	Length, km
1	0.10	40.00	40	0.10
2	1.35	40.00	40	1.35
3	0.90	38.76	30	0.05
			40	0.85
4	1.15	38.72	30	0.06
			40	1.09
5	1.45	37.75	30	0.15
			40	1.30
6	0.45	36.25	30	0.09
			40	0.36
7	0.85	33.85	30	0.35
			40	0.50
8	0.85	32.76	30	0.46
			40	0.39
9	0.80	31.53	30	0.57
			40	0.23
10	0.95	25.01	24	0.69
			30	0.26
11	1.20	23.01	20	0.19
			24	1.01
12	3.50	20.15	20	3.29
			24	0.21
13	0.80	19.04	16	0.11
			20	0.69
14	0.50	14.48	12	0.10
			16	0.40
15	0.55	12.00	12	0.55
16	2.73	19.91	16	0.03
			20	2.70
17	1.75	23.05	20	0.26
			24	1.49
18	0.80	26.57	24	0.33
			30	0.47
19	0.40	26.82	24	0.15
			30	0.25
20	2.20	35.22	30	0.62
			40	1.58
21	1.50	16.37	16	1.26
			20	0.24
22	0.50	12.00	12	0.50
23	2.65	29.51	24	0.11
			30	2.54
24	1.23	19.33	16	0.11
			20	1.12
25	1.30	16.42	16	1.07
			20	0.23
26	0.85	12.00	12	0.85
27	0.30	20.49	20	0.24
			24	0.06
28	0.75	22.41	20	0.21
			24	0.54
29	1.50	18.86	16	0.25
			20	1.25
30	2.00	17.12	16	1.16
			20	0.84
31	1.60	14.56	12	0.30
			16	1.30
32	0.15	12.00	12	0.15
33	0.86	12.00	12	0.86
34	0.95	19.55	16	0.06
			20	0.89

where $r(1, k)$ is a directed path connecting the source nodes 1 and k.

Pumping at Demand Nodes and Links

If additional pumping facilities are considered at the demand nodes as well, the objective function remains (30),

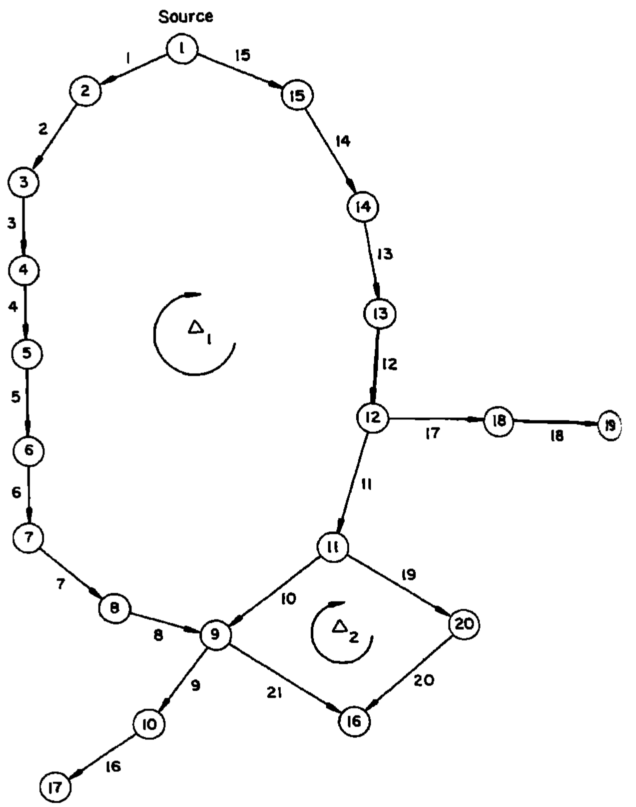

Fig. 3. Layout for New York City water supply system.

with the last two summations taken over all the source and demand nodes. Moreover, the hydraulic head requirement equation (4) will be modified for each demand node k as

$$\sum_{i \in r(k)} \pm J_i \leqq H_1 + h_1 + H_k - h_k^{\min} \tag{32}$$

TABLE 4. Node Data of New York City Water Supply Network

Node	Minimum Head Required, ft	Optimal Head Obtained, ft	Demand, $ft^3\ s^{-1}$
1	300.0	300.0	−2017.5
2	255.0	294.4	92.4
3	255.0	286.7	92.4
4	255.0	284.4	88.2
5	255.0	282.4	88.2
6	255.0	281.0	88.2
7	255.0	278.6	88.2
8	255.0	276.1	88.2
9	255.0	273.6	170.0
10	255.0	273.6	1.0
11	255.0	273.7	170.0
12	255.0	275.0	117.1
13	255.0	278.0	117.1
14	255.0	285.5	92.4
15	255.0	293.3	92.4
16	260.0	260.0	170.0
17	272.8	272.8	57.5
18	255.0	262.5	117.1
19	255.0	255.0	117.1
20	255.0	264.5	170.0

TABLE 5. Link Data of New York City Water Distribution Network

Link	From Node	To Node	Length, ft	Existing Diameter, inches	Initial Flow, $ft^3\ s^{-1}$	Optimal Flow, $ft^3\ s^{-1}$	New Diameter, inches
1	1	2	11,600	180	873.7	872.4	0.00
2	2	3	19,800	180	781.3	780.0	0.00
3	3	4	7,300	180	688.9	687.6	0.00
4	4	5	8,300	180	600.7	599.4	0.00
5	5	6	8,600	180	492.5	491.2	0.00
6	6	7	19,100	180	424.3	423.0	0.00
7	7	8	9,600	132	335.3	334.8	0.00
8	8	9	12,500	132	247.9	246.6	73.62
9	9	10	9,600	180	58.5	58.5	0.00
10	11	9	11,200	204	122.8	129.3	0.00
11	12	11	14,500	204	490.6	491.9	0.00
12	13	12	12,200	204	841.9	483.2	0.00
13	14	13	24,100	204	959.0	960.3	0.00
14	15	14	21,100	204	1051.4	1052.7	0.00
15	1	15	15,500	204	1143.8	1145.1	0.00
16	10	17	26,400	72	57.5	57.5	99.01
17	12	18	31,200	72	234.2	234.2	98.75
18	18	19	24,000	60	117.1	117.1	78.97
19	11	20	14,400	60	197.8	192.6	83.82
20	20	16	38,400	60	27.8	22.6	0.00
21	9	16	26,400	72	142.2	147.4	66.59

If additional pumping facilities are also considered at links, the following pumping costs are added to the objective function (30)

$$b_4 \sum_{i=1}^{t} q_i^{e_1'} H_{(i)}^{e_2'} + b_5 \sum_{i=1}^{t} q_i H_{(i)} \tag{33}$$

where $H_{(i)}$ is the pumping pressure at the pump station in link i ($i = 1, \cdots, t$), and b_4, b_5 are empirical constants. Moreover, the head balance equation (3) along each loop is replaced by

$$\sum_{i \in \text{loop } p} \pm J_i = \sum_{i \in \text{loop } p} \pm H_{(i)} \tag{34}$$

and the hydraulic head requirement equation (4) for each node k is replaced by

$$\sum_{i \in r(k)} \pm J_i \leqq H_1 + h_1 + \sum_{i \in r(k)} \pm H_{(i)} + H_k - h_k^{\min} \tag{35}$$

TABLE 6. Comparative Study of Numerical Results on New York City Water Distribution Network

Link	Length, m	Diameter of New Pipes, inches				
		Shaake and Lai [1969]	*Quindry et al.* [1981]	*Bhave* [1985]	*Morgan and Goulter* [1985]	O. Fujiwara and D. B. Khang (presented paper)
1	11,600	52.02	0.00	0.00	0.00	0.00
2	19,800	49.90	0.00	0.00	0.00	0.00
3	7,300	63.41	0.00	0.00	0.00	0.00
4	8,300	55.59	0.00	0.00	0.00	0.00
5	8,600	57.25	0.00	0.00	0.00	0.00
6	19,100	59.19	0.00	0.00	0.00	0.00
7	9,600	59.06	0.00	0.00	144.00	73.62
8	12,500	54.95	0.00	0.00	0.00	0.00
9	9,600	0.00	0.00	0.00	0.00	0.00
10	11,200	0.00	0.00	0.00	0.00	0.00
11	14,500	116.21	119.02	0.00	0.00	0.00
12	12,200	125.25	134.39	0.00	0.00	0.00
13	24,100	126.87	132.49	0.00	0.00	0.00
14	21,100	133.07	132.87	0.00	0.00	0.00
15	15,500	126.52	131.37	136.43	0.00	0.00
16	26,400	19.52	19.26	87.37	96.00	99.01
17	31,200	91.83	91.71	99.23	96.00	98.75
18	24,000	72.76	72.76	78.17	84.00	78.97
19	14,400	72.61	72.64	54.40	60.00	83.82
20	38,400	0.00	0.00	0.00	0.00	0.00
21	26,400	54.82	54.97	81.50	84.00	66.59
Total cost, \$million		77.611	63.581	40.180	39.229	36.138

TABLE 7. Comparing Split Pipe Solutions

	Morgan and Goulter Solution		Fujiwara and Khang Solution	
Link	Diameter, inches	Length, m	Diameter, inches	Length, m
7	144	9,600	72	7,737
			84	1,863
16	96	23,432	96	17,957
	108	2,968	108	8,443
17	96	31,200	96	22,006
			108	9,194
18	72	58	72	7,521
	84	23,942	84	16,479
19	48	4,528	72	137
	60	9,872	84	14,263
21	72	3,493	60	8,549
	84	22,907	72	1,786

Toal cost for the Morgan and Goulter solution is 38.9 million dollars and for the Fujiwara and Khang solution is 36.6 million dollars.

where J_i and $H_{(i)}$ take the same sign. Note that the new objective function is still convex in J but not concave in q and $H = [H_1, \cdots, H_z, H_{(1)}, \cdots, H_{(t)}]$. However, since all the modified constraints remain linear, efficient local optimal solution techniques can be still applicable.

Multiple Loadings

To deal with multiple loading patterns $a_k(l)$ $(l = 1, \cdots, u)$, parameterized variables such as pump heads $H_k(l)$ and $H_{(i)}(l)$, link flows $q_i(l)$ and link head loss $J_i(l)$ at each loading pattern l can be introduced. The pipe cost in the objective function (30) remains the same, and the pump capital cost is still

$$b_2 \sum_{k=1}^{z} |a_k|^{e_1} H_k^{e_2} + b_4 \sum_{i=1}^{t} q_i^{e_1} H_{(i)}^{e_2}$$

however, a_k is now taken as $|a_k| = \max_l |a_k(l)|$. Furthermore, the following new constraints are added:

$$H_k \geqq H_k(l) \qquad \text{for all } k \text{ and } l \qquad (36)$$

$$q_i \geqq q_i(l) \qquad \text{for all } i \text{ and } l \qquad (37)$$

$$H_{(i)} \geqq H_{(i)}(l) \qquad \text{for all } i \text{ and } l \qquad (38)$$

The pump operation cost is replaced by the weighted average cost of each loading pattern as follows:

$$\sum_l w_l[b_3 \sum_k |a_k(l)| H_k(l) + b_5 \sum_i q_i(l) H_{(i)}(l)] \qquad (39)$$

where $w_l \geqq 0$ and $\Sigma w_l = 1$.

The modification of the constraints are straightforward.

New Expansion

The two models proposed in this paper can be combined into a mixed optimization problem involving both the parallel expansion of an existing network and the installation of new nodes and links. In this case, equations (24), (26), (27), (28), and (29) should be used for link i along which parallel expansion is considered, while equations (7), (8), (9), (10), and (11) should be used for the new link i.

CONCLUDING REMARKS

The solution methods for the optimal design of water distribution networks available in the literature terminate after finding a local optimal solution or its approximate. The method proposed in this paper, on the other hand, tries to find better and better local optima to the original problem. The first phase of the algorithm allows improving a given solution to a local optimal solution by a gradient approach. The concave program in the second phase then makes it possible to shift the obtained local optimum to an improved one which may be located far away and which still be used as initial value to restart the first phase to obtain a new and better local optimal solution. The procedure continues until no further improvement can be achieved.

Since the subproblems in both the two phases are of a much smaller size than that of the original problem and their structure is much simpler (being a minimization of a convex or concave objective function over a polyhedral set), the method is expected to be computationally more efficient than the direct application of general purpose NLP codes (such as MINOS or GRG2).

In this paper the direction of link flows are preassigned in the model so that q_i and J_i are positive. However, the NLPG method can be carried out to incorporate the change of link flow direction in the following way: if in the local optimal solution obtained the lower bound for the flow is met at some link and the gradient at that point indicates that further reduction in the flow may lower the system cost, then the direction of this link is reversed and the NLPG method is reapplied to this new configuration. The new solution then replaces the incumbent one if it provides better cost value. The whole process is repeated until no reduction can be achieved in the total system cost.

APPENDIX

The local optimality of the solution obtained by the first phase follows immediately from the general theorem below

Theorem: Consider the NLP problem (P):

$$\min_{x,y} f(x, y)$$

subject to

$$Ax \leqq a$$

$$By \leqq b$$

$$h(x, y) \leqq 0$$

where f is a real valued function and $h = [h_1, \cdots, h_p]$ is a vector-valued function defined for $x \in R^m$ and $y \in R^n$, A and B are k by m and l by n matrices, respectively, while a and b are k and l vectors, respectively. Assume that f is concave in x and convex in y and h is linear in x and in y. In addition, suppose that $\bar{x}$ is a local optimal solution of the problem $\min_x \{\phi(x): Ax \leqq a\}$, where $\phi(x)$ is the optimal objective value of the subproblem $(P(x))$

$$\phi(x) = \min_y f(x,y)$$

subject to

$$By \leqq b$$

$$h(x, y) \leqq 0$$

for a fixed x and suppose that $\bar{y}$ is a solution of $P(\bar{x})$. Then $(\bar{x}, \bar{y})$ is a local optimal solution of the problem (P).

Proof. Since $\bar{x}$ is a local minimum of $\phi(x)$ over $Ax \leqq a$, there exists a sufficiently small positive number ε such that $\phi(\bar{x}) \leqq \phi(x)$ for every x satisfying $Ax \leqq a$ and $\|x - \bar{x}\| \leqq \varepsilon$. On the other hand, since $P(x)$ is a convex program, each of its local solutions is also a global one and hence $f(x, y) \geqq \phi(x)$ for every y satisfying $By \leqq b$ and $h(x, y) \leqq 0$. Thus for every feasible solution (x, y) of (P) such that $\|x - \bar{x}\| \leqq \varepsilon$ we have

$$f(x, y) \geqq \phi(x) \geqq \phi(\bar{x}) \geqq f(\bar{x}, \bar{y})$$

which proves that $(\bar{x}, \bar{y})$ is a local optimum of (P).

NOTATION

- a_k water demand at node k.
- d_i diameter of existing pipe along link i.
- d_{max} upper bound for pipe diameters.
- d_{min} lower bound for pipe diameters.
- h_1 initial hydraulic head at source node 1.
- H_1 head lift provided by pumping at source node 1.
- J_i headloss along link i.
- L_i length of link i.
- m number of independent loops.
- n number of nodes.
- q_i flow rate along link i.
- q_i^{min} minimum required flow rate along link i.
- t number of links.
- x_i diameter of new pipe along link i.

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REFERENCES

Alperovits, E., and U. Shamir, Design of optimal water distribution system, *Water Resour. Res.*, *13*(6), 885–900, 1977.

Berge, C., *Graphs*, pp. 15–21, North-Holland, Amsterdam, 1985.

Bhave, P. R., Noncomputer optimization of single-source networks, *J. Environ. Eng. Div. Am. Soc. Civ. Eng.*, *104*(EE4), 799–814, 1978.

Bhave, P. R., Optimization of gravity-fed water distribution systems: Theory, *J. Environ. Eng. Div. Am. Soc. Civ. Eng.*, *109*(EE1), 189–205, 1983.

Bhave, P. R., Optimal expansion of water distribution systems, *J. Environ. Eng. Div. Am. Soc. Civ. Eng.*, *111*(EE2), 177–197, 1985.

Dennis, J. E., and R. B. Schnabel, *Numerical Methods for Unconstrained Optimization and Nonlinear Equations*, pp. 208–212, Prentice Hall, Englewood Cliffs, N. J., 1983.

Falk, J., and K. R. Hoffman, A successive underestimation method for concave minimization problems, *Math. Oper. Res.*, *1*(2), 251–259, 1976.

Fiacco, A. V., *Introduction to Sensitivity and Stability Analysis in Nonlinear Programming*, pp. 72–90, Academic, San Diego, Calif., 1983.

Fletcher, R., *Practical Methods for Optimization*, 2nd ed., pp. 240–245, John Wiley, New York, 1987.

Fujiwara, O., B. Jenchaimahakoon, and N. C. P. Edirisinghe, A modified LPG method for optimal design of looped water distribution networks, *Water Resour. Res.*, *23*(6), 977–982, 1987.

Gill, P. E., W. Murray, and M. H. Wright, *Practical Optimization*, pp. 167–175, Academic, San Diego, Calif., 1981.

Jacoby, S. L. S., Design of optimal hydraulic networks, *J. Hydraul. Eng.*, *94*(HY3), 641–661, 1968.

Mahjoub, Z., Contribution a l'etude de l'optimisation des reseaux mailles, pp. 51–142, These (Docteur d'Etat), l'Inst. Natl. Polytech. de Toulouse, Toulouse, France, 1983.

Morgan, D. R., and I. C. Goulter, Optimal urban water distribution design, *Water Resour. Res.*, *21*(5), 642–652, 1985.

Pitchai, R., A model for designing water distribution pipe networks, Ph.D. thesis, Harvard Univ., Cambridge, Mass., 1966.

Quindry, G. E., E. D. Brill, J. C. Liebman, and A. Robinson, Comments on "Design of optimal water distribution systems" by E. Alperovits and U. Shamir, *Water Resour. Res.*, *15*(6), 1651–1654, 1979.

Quindry, G. E., E. D. Brill, and J. C. Liebman, Optimization of looped water distribution systems, *J. Environ. Eng. Div. Am. Soc. Civ. Eng.*, *107*(EE4), 665–679, 1981.

Rosen, J., The gradient projection method for nonlinear programming, I, Linear constraints, *J. Soc. Ind. Appl. Math.*, *8*, 514–532, 1960.

Rowell, W. F., and W. J. Barnes, Obtaining layout of water distribution systems, *J. Hydraul. Eng.*, *108*(HY1), 137–148, 1982.

Schaake, J. C., Jr., and D. Lai, Linear programming and dynamic programming application of water distribution network design, *Rep. 116*, Hydrodyn. Lab., Dep. of Civ. Eng., MIT Press, Cambridge, Mass., 1969.

Shamir, U., Minimum cost design of water distribution networks, Report of Civil Engineering, Mass. Inst. of Technol., Cambridge, Mass., 1964.

Shamir, U., Optimal design and operation of water distribution systems, *Water Resour. Res.*, *11*(4), 27–36, 1974.

Shamir, U., and C. D. D. Howard, Water distribution system analysis, *J. Hydraul. Eng.*, *94*(HY1), 219–234, 1968.

Swamee, P. K., V. Kumar, and P. Khanna, Optimization of dead end water distribution systems, *J. Environ. Div. Am. Soc. Civ. Eng.*, *99*(EE2), 123–134, 1973.

Thieu, T. V., B. T. Tam, and V. T. Ban, An outer approximation method for globally minimizing a concave function over a compact convex set, *Acta Math. Vietnamica*, *8*(1), 21–40, 1983.

Tuy, H., Concave programming under linear constraints, *Dokl. Akad. Nauk SSSR*, *159*(1), 32–35, 1964. (*Soviet Math.*, Engl. Transl., *5*(8), 1437–1440, 1964.)

Tuy, H., On outer approximation methods for solving concave minimization problems, *Acta Math. Vietnamica*, *8*(2), 3–34, 1983.

Tuy, H., Vetex-facet duality and polyhedral annexation method for concave minimization, *Res. Pap. 159*, Ind. Eng. and Manage. Div., Asian Inst. of Technol., Bangkok, Thailand, 1987.

Watanatada, T., Least cost design of water distribution systems, *J. Hydraul. Eng.*, *99*(HY9), 1497–1513, 1973.

Zwart, P. B., Global maximization of a convex function with linear inequality constraints, *Oper. Res.*, *22*(3), 602–609, 1974.

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